

Unit 10

Determinants

Unit Goals

- Be able to define the determinant.
- Be able to manipulate algebraic expressions of determinants.
- Be able compute determinants of 2-by-2 and 3-by-3 matrices using direct methods.
- Be able to compute determinants of n -by- n matrices by recursion.
- Be able to compute determinants of n -by- n matrices by LU decomposition.
- Be able to use the determinant to assess the invertibility of a matrix.

Unit 10.1

Defining the determinant.

What is a determinant?

Definition:

- The determinant of a square matrix is an algebraic expression (in terms of the elements of the matrix) that can be used to determine whether the matrix has an inverse.
- If a matrix has a determinant of 0, it is singular; the inverse does not exist; if the determinant is non-zero, the matrix is nonsingular.
- Just as the matrix inverse is only defined for square matrices, the determinant is only defined for square matrices.

The 2-by-2 case.

- The determinant of a square matrix is an algebraic expression (in terms of the elements of the matrix) that can be used to determine whether or not the matrix has an inverse.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} a & b \\ 0 & d - \frac{bc}{a} \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & \frac{ad - bc}{a} \end{bmatrix}, a \neq 0$$

- Inverse exists if and only if:

$$ad - bc \neq 0$$

$$\det(A) = |A| = (ad - bc)$$

The 3-by-3 case.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} - \frac{a_{12}a_{21}}{a_{11}} & a_{23} - \frac{a_{13}a_{21}}{a_{11}} \\ 0 & a_{32} - \frac{a_{12}a_{31}}{a_{11}} & a_{33} - \frac{a_{13}a_{31}}{a_{11}} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}} & \frac{a_{23}a_{11} - a_{13}a_{21}}{a_{11}} \\ 0 & \frac{a_{32}a_{11} - a_{12}a_{31}}{a_{11}} & \frac{a_{33}a_{11} - a_{13}a_{31}}{a_{11}} \end{bmatrix}, a_{11} \neq 0$$

$$\rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}} & \frac{a_{23}a_{11} - a_{13}a_{21}}{a_{11}} \\ 0 & 0 & \frac{a_{33}a_{11} - a_{13}a_{31}}{a_{11}} - \left(\frac{a_{23}a_{11} - a_{13}a_{21}}{a_{11}} \right) \left(\frac{a_{32}a_{11} - a_{12}a_{31}}{a_{22}a_{11} - a_{12}a_{21}} \right) \end{bmatrix}, a_{22}a_{11} - a_{12}a_{21} \neq 0$$

$$= \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & \frac{a_{22}a_{11} - a_{12}a_{21}}{a_{11}} & a_{23} \\ 0 & 0 & \frac{(a_{33}a_{11} - a_{13}a_{31})(a_{22}a_{11} - a_{12}a_{21}) - (a_{23}a_{11} - a_{13}a_{21})(a_{32}a_{11} - a_{12}a_{31})}{a_{11}(a_{22}a_{11} - a_{12}a_{21})} \end{bmatrix}$$

The 3-by-3 case.

$$\begin{aligned}
 & \frac{(a_{33}a_{11} - a_{13}a_{31})(a_{22}a_{11} - a_{12}a_{21}) - (a_{23}a_{11} - a_{13}a_{21})(a_{32}a_{11} - a_{12}a_{31})}{a_{11}(a_{22}a_{11} - a_{12}a_{21})} \\
 & \quad \downarrow \\
 & \frac{a_{33}a_{11}a_{22}a_{11} - a_{33}a_{11}a_{12}a_{21} - a_{13}a_{31}a_{22}a_{11} + a_{13}a_{31}a_{12}a_{21} - a_{23}a_{11}a_{32}a_{11} + a_{23}a_{11}a_{12}a_{31} + a_{13}a_{21}a_{32}a_{11} - a_{13}a_{21}a_{12}a_{31}}{a_{11}(a_{22}a_{11} - a_{12}a_{21})} \\
 & \quad \downarrow \\
 & \frac{a_{33}a_{11}a_{22}a_{11} - a_{33}a_{11}a_{12}a_{21} - a_{13}a_{31}a_{22}a_{11} + a_{13}a_{31}a_{12}a_{21} - a_{23}a_{11}a_{32}a_{11} + a_{23}a_{11}a_{12}a_{31} + a_{13}a_{21}a_{32}a_{11} - a_{13}a_{21}a_{12}a_{31}}{a_{11}(a_{22}a_{11} - a_{12}a_{21})} \\
 & \quad \downarrow \\
 & \frac{a_{33}a_{11}a_{22} - a_{33}a_{12}a_{21} - a_{13}a_{31}a_{22} - a_{23}a_{32}a_{11} + a_{23}a_{12}a_{31} + a_{13}a_{21}a_{32}}{(a_{22}a_{11} - a_{12}a_{21})}
 \end{aligned}$$

- Inverse exists if and only if:

$$a_{33}a_{11}a_{22} - a_{33}a_{12}a_{21} - a_{13}a_{31}a_{22} - a_{23}a_{32}a_{11} + a_{23}a_{12}a_{31} + a_{13}a_{21}a_{32} \neq 0$$

$$\det(A) = |A| = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} + a_{12}a_{23}a_{31} - a_{12}a_{21}a_{33} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$$

The triangular case.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

- Inverse exists if and only if all diagonal entries are non-zero.

$$a_{11}, a_{22}, a_{33}, \cdots, a_{nn} \neq 0$$

$$\det(A) = |A| = a_{11}a_{22}a_{33} \cdots a_{nn} = \sum_{i=1}^n a_{ii}$$

Properties of the determinant.

- The determinant of the identity matrix is one:

$$\det(I) = |I| = 1$$

- The determinant of any singular matrix is zero:

$$\det(A) = |A| = 0 \iff \nexists A^{-1}$$

- Any matrix with identical rows.
- Any matrix with identical columns.
- Any matrix with a zero row or a zero column.

Properties of the determinant

- Exchanging rows changes the sign of the determinant:

$$A \xrightarrow{R_i \leftrightarrow R_j} B \Rightarrow |B| = -|A|$$

- Scaling a row scales the determinant by the same amount:

$$A \xrightarrow{kR_i \rightarrow R_i} B \Rightarrow |B| = k|A|$$

- Adding a scalar multiple of one row to another does not change the determinant :

$$A \xrightarrow{R_i + kR_j \rightarrow R_i} B \Rightarrow |B| = |A|$$

Properties of the determinant:

- The determinant of a product of matrices is the product of the determinants:

$$C = AB \Rightarrow |C| = |A||B|$$

- The determinant of the inverse is the inverse of the determinant:

$$|A^{-1}| = (|A|)^{-1} = \frac{1}{|A|}$$

- A matrix has the same determinant as its transpose:

$$|A| = |A^T|$$

Determinants and elimination.

- Consider Gaussian elimination as represented by the LU decomposition:

$$A \xrightarrow{P} A' \xrightarrow{L^{-1}} U, \quad PA = LU$$

$$|P||A| = |L||U| \rightarrow |A| = \frac{|L||U|}{|P|}$$

- P is an identity matrix with rearranged rows, and L is a lower triangular matrix with ones on the diagonal:

$$|P| = \pm 1, |L| = 1 \rightarrow |A| = \pm |U|$$

Determinants and elimination.

The determinant of any matrix can be found using elimination:

- Perform forward elimination, without row scaling, to an echelon form:

$$A \rightarrow U$$

- The product of the diagonal entries the echelon form gives the determinant of the original matrix to within a sign:

$$|A| = \pm |U|$$

- Count the row exchanges: if there were an even number, leave the sign unchanged, if there were an odd number change the sign.

Determinants and elimination.

Example: $\begin{vmatrix} 1 & 2 & 1 & 2 \\ 2 & 2 & 2 & 1 \\ 1 & 0 & 1 & -2 \\ -1 & 2 & 2 & 4 \end{vmatrix} = ?$

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 2 & 2 & 2 & 1 \\ 1 & 0 & 1 & -2 \\ -1 & 2 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -2 & 0 & -3 \\ 0 & -2 & 0 & -4 \\ 0 & 4 & 3 & 6 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -2 & 0 & -3 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 3 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & -2 & 0 & -3 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} = U$$

- We used one row exchange (odd):

$$|U| = (1)(-2)(3)(-1) = +6, \quad |A| = -|U| = -6$$

Unit 10.1.1

Worked examples using the properties of the determinant.

Find the determinant

Find the determinant:

$$|L|, \quad L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 \\ 4 & -2 & 4 & 0 \\ 5 & -3 & 5 & -3 \end{bmatrix}$$

Find the determinant

$$L = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 \\ 4 & -2 & 4 & 0 \\ 5 & -3 & 5 & -3 \end{bmatrix}$$

- Lower triangular matrix, simply take the product of the diagonal elements:

$$|L| = (2)(-1)(4)(-3) = 24$$

Find the determinant

Find the determinant:

$$|P|, \quad P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Find the determinant

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Three (odd) row exchanges to get the identity matrix:

$$|P| = -|I| = -1$$

Find the determinant

Find the determinant:

$$|A|, \quad A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

Find the determinant

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & -1/2 & 3 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 0 \\ 0 & -2 & -1 & 0 \\ 0 & 0 & -1/2 & 3 \\ 0 & 0 & 0 & 16 \end{bmatrix} = U$$

- Elimination without row exchange:

$$|A| = |U| = (1)(-2) \left(\frac{-1}{2} \right) (16) = 16$$

Find the determinant

Find the determinant:

$$|A|, \quad A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 1 \\ 2 & 1 & 2 & 3 \end{bmatrix}$$

Find the determinant

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 1 \\ 2 & 1 & 2 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$
$$\rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1/2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1/2 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = U$$

- We used two row exchanges (even):

$$|U| = (1)(2) \left(\frac{1}{2}\right) (1) = 1, \quad |A| = |U| = 1$$

Unit 10.2

Using the determinant.

Direct forms for the determinant.

- The determinant of any matrix can be found using elimination.

- Are there more direct definitions?

- 2x2: $|A| = (a_{11}a_{22} - a_{12}a_{21})$

- 3x3:
$$|A| = +a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$

- Sum of terms, each a product of n elements. First indices follow 1, 2, ... , n; second indices are “scrambled.”

Leibniz formula: Permutations

Formal definition:

$$\det(A) = |A| = \sum_{\sigma \in S_n} \left(\text{sgn}(\sigma) \prod_{i=1}^n a_{i\sigma(i)} \right)$$

- S_n : The set of all possible permutations of the sequence 1, 2, ... n.
- σ : One particular permutation (re-ordering) of the sequence 1, 2, ... n.
- $\sigma(i)$: The i^{th} element of a particular permutation.
- $\text{sgn}(\sigma)$: The *parity* of the permutation. $\text{sgn}(\sigma) = (-1)^p$ where p is number of pairwise swaps to required to convert the permutation into the original sequence.

Elegant, but not practical, especially for “by-hand” calculations.

Direct forms for the determinant.

- 2x2: $|A| = (a_{11}a_{22} - a_{12}a_{21})$

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}|a_{22}| - a_{12}|a_{21}|$$

- 3x3: $|A| = +a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$
 $-a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

- A sum of n determinants of (n-1)-by-(n-1) matrices.

Direct forms for the determinant.

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \rightarrow \begin{vmatrix} a_{11} & & \\ & a_{22} & a_{23} \\ & a_{32} & a_{33} \end{vmatrix}, \begin{vmatrix} & a_{12} & \\ a_{21} & & a_{23} \\ a_{31} & & a_{33} \end{vmatrix}, \begin{vmatrix} & & a_{13} \\ a_{21} & a_{22} & \\ a_{31} & a_{32} & \end{vmatrix}$$

- Each term includes one element of the first row, and the determinant of the matrix left after deleting the row and column of the coefficient.

Laplace expansion: Cofactors.

Expansion on row i:

$$|A| = \sum_{j=1}^n a_{ij} C_{ij}, \quad C_{ij} = (-1)^{i+j} |M_{ij}|$$

Expansion on column j:

$$|A| = \sum_{i=1}^n a_{ij} C_{ij}, \quad C_{ij} = (-1)^{i+j} |M_{ij}|$$

- M_{ij} is the matrix obtained when row i and column j are removed from A .
- $|M_{ij}|$ is called the ij^{th} **minor** of A .
- C_{ij} is the called the ij^{th} **cofactor** of A .

Recursive use of Laplace expansion

$$\begin{aligned}
 & \begin{vmatrix} 1 & 2 & 0 & 0 & 0 \\ 2 & 1 & 2 & 0 & 0 \\ 0 & 2 & 1 & 2 & 0 \\ 0 & 0 & 2 & 1 & 2 \\ 0 & 0 & 0 & 2 & 1 \end{vmatrix} = 1 \begin{vmatrix} 1 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 2 & 1 \end{vmatrix} \\
 &= 1 \left(1 \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix} \right) - 2 \left(2 \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix} \right) = -3 \begin{vmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{vmatrix} \\
 &= -3 \left(1 \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix} \right) - 2 \left(2 \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} \right) = -7 \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} + 6 \begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix} \\
 &= -7(1 - 4) + 6(2 - 0) = 33
 \end{aligned}$$

What use is the determinant?

- The determinant gives an algebraic expression that can tell us when a matrix is invertible.
- This can be useful when a matrix contains symbolic constants, and we would like to understand under what conditions the matrix will be singular.
- We will use this idea when we discuss eigenvalues!

What use is the determinant?

Inverse from classical adjoints:

- For any non-singular matrix A :

$$A^{-1} = |A|^{-1} \text{adj}(A)$$

- The classical adjoint (or adjugate) of A is the transpose of the matrix of cofactors:

$$\text{adj}(A) = C^T$$

$$C = [c_{ij}], \quad c_{ij} = (-1)^{(i+j)} |M_{ij}|$$

What use is the determinant?

$$A^{-1} = |A|^{-1} \text{adj}(A), \text{adj}(A) = C^T, C = [c_{ij}], c_{ij} = (-1)^{(i+j)} |M_{ij}|$$

• **2x2:** $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$|A| = ad - bc$$

$$c_{11} = d, c_{12} = -c, c_{21} = -b, c_{22} = a \rightarrow C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$$

$$A^{-1} = |A|^{-1} C^T = \left(\frac{1}{ad - bc} \right) \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

What use is the determinant?

- **3x3:** $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, A^{-1} = |A|^{-1} C^T$

$$|A| = +a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} \\ - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$

$$c_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}, c_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}, \dots, c_{33} = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}$$

- **Nine 2x2 determinants! Not an efficient method.**
- **This is most useful for matrices with symbolic constants, rather than simply numbers.**

What use is the determinant?

Cramer's Rule:

- Given any non-singular square matrix A , we can directly write the solution for any right-hand side as:

$$A\vec{x} = \vec{b} \rightarrow x_i = \frac{|B_i|}{|A|}$$

- B_i is the matrix obtained by replacing column i of A with $\vec{b}$:

$$\begin{aligned} A &= [\vec{a}_1 & \cdots & \vec{a}_{i-1} & \vec{a}_i & \vec{a}_{i+1} & \cdots & \vec{a}_n] \\ B_i &= [\vec{a}_1 & \cdots & \vec{a}_{i-1} & \vec{b} & \vec{a}_{i+1} & \cdots & \vec{a}_n] \end{aligned}$$

What use is the determinant?

$$A\vec{x} = \vec{b} \rightarrow x_i = \frac{|B_i|}{|A|}$$

• **2x2:**
$$\begin{array}{rcl} ax & +by & = k \\ cx & +dy & = l \end{array} \rightarrow A\vec{x} = \vec{b}, A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \vec{b} = \begin{bmatrix} k \\ l \end{bmatrix}$$

$$|A| = ad - bc$$

$$|B_1| = \begin{vmatrix} k & b \\ l & d \end{vmatrix} = kd - bl, \quad |B_2| = \begin{vmatrix} a & k \\ c & l \end{vmatrix} = al - ck$$

$$x = x_1 = \frac{|B_1|}{|A|} = \frac{kd - bl}{ad - bc}, \quad y = x_2 = \frac{|B_2|}{|A|} = \frac{al - ck}{ad - bc}$$

What use is the determinant?

- **3x3:** $A\vec{x} = \vec{b}, A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}, x_i = \frac{|B_i|}{|A|}$

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}, |B_1| = \begin{vmatrix} b_1 & a_{12} & a_{13} \\ b_2 & a_{22} & a_{23} \\ b_3 & a_{32} & a_{33} \end{vmatrix}, |B_2| = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}, |B_3| = \begin{vmatrix} a_{11} & a_{12} & b_1 \\ a_{21} & a_{22} & b_2 \\ a_{31} & a_{32} & b_3 \end{vmatrix}$$

- **Four 3x3 determinants! Not an efficient method.**
- **Again, this is most useful for systems with symbolic constants, rather than simply numbers.**

What use is the determinant?

Cramer's Rule:

- Given any non-singular square matrix A , we can directly write the solution for any right-hand side as:

$$A\vec{x} = \vec{b} \rightarrow x_i = \frac{|B_i|}{|A|}$$

- For an $n \times n$ matrix, we need to compute $n + 1$ determinants, each with $n!$ terms of n elements:

$$O(n^2 n!)$$

What is the use of determinant?

- The determinant provides a tool~~s~~ that allows us to write analytical expressions for:
 - The singularity of a matrix.
 - The inverse of nonsingular matrix.
 - The solution to a system with a nonsingular coefficient matrix.
- These methods are not well-suited to solving general systems:
 - Elimination is much faster for all but the smallest systems.
 - But ... can be useful when symbolic constants are involved!

Unit 10.2.1

**Worked examples using the
determinant.**

Conditions on singularity.

- For what values of a , b , c and d is the following matrix singular?

$$A = \begin{bmatrix} 1 & a & 1 \\ b & 1 & c \\ 1 & d & 1 \end{bmatrix}$$

Conditions on singularity.

$$\begin{vmatrix} 1 & a & 1 \\ b & 1 & c \\ 1 & d & 1 \end{vmatrix} = 1 \begin{vmatrix} 1 & c \\ d & 1 \end{vmatrix} - a \begin{vmatrix} b & c \\ 1 & 1 \end{vmatrix} + 1 \begin{vmatrix} b & 1 \\ 1 & d \end{vmatrix}$$

$$= (1 - cd) - a(b - c) + (bd - 1)$$

$$= 1 - cd - ab + ac + bd - 1$$

$$= a(c - b) + d(b - c) = (a - d)(c - b)$$

$$|A| = 0 \rightarrow (a - d)(c - d) = 0 \rightarrow a = d \text{ or } b = c$$

- **A will be singular if (and only if) either $a = d$ or $b = c$.**

Cramer's Rule

- Find a general form for the solution of the following system of equations.

$$x_1 = \quad \quad \quad ax_2 \quad + ax_3 \quad + b$$

$$x_2 = \quad ax_1 \quad \quad \quad + ax_3 \quad + b$$

$$x_3 = \quad ax_1 \quad + ax_2 \quad \quad \quad + b$$

- Recall:** $x_i = \frac{|B_i|}{|A|}$, $B_i = \begin{bmatrix} \vec{a}_1 & \cdots & \vec{a}_{i-1} & \vec{b} & \vec{a}_{i+1} & \cdots & \vec{a}_n \end{bmatrix}$

Cramer's Rule

- Recast in matrix form:

$$\begin{aligned}x_1 &= && ax_2 & +ax_3 & +b \\x_2 &= & ax_1 & & +ax_3 & +b \\x_3 &= & ax_1 & +ax_2 & & +b\end{aligned} \rightarrow \vec{x} = A\vec{x} + \vec{b} \rightarrow (I - A)\vec{x} = \vec{b}$$

$$A = \begin{bmatrix} 0 & a & a \\ a & 0 & a \\ a & a & 0 \end{bmatrix}, M = (I - A) = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}, x_i = \frac{|B_i|}{|M|}, B_i = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{b} \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$

$$\begin{aligned} |M| &= 1 \begin{vmatrix} 1 & -a \\ -a & 1 \end{vmatrix} + a \begin{vmatrix} -a & -a \\ -a & 1 \end{vmatrix} - a \begin{vmatrix} -a & 1 \\ -a & -a \end{vmatrix} \\ &= (1 - a^2) + a(-a - a^2) - a(a^2 + a) \\ &= 1 - a^2 - a^2 - a^3 - a^3 - a^2 \\ &= 1 - 3a^2 - 2a^3 \\ &= (1 + a)(1 - a - 2a^2) \\ &= (1 + a)(1 + a)(1 - 2a) \\ &= (1 + a)^2(1 - 2a) \end{aligned}$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}, x_i = \frac{|B_i|}{|M|}, B_i = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{b} \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$

$$\begin{aligned} |B_1| &= \begin{vmatrix} b & -a & -a \\ b & 1 & -a \\ b & -a & 1 \end{vmatrix} = b \begin{vmatrix} 1 & -a \\ -a & 1 \end{vmatrix} + a \begin{vmatrix} b & -a \\ b & 1 \end{vmatrix} - a \begin{vmatrix} b & 1 \\ b & -a \end{vmatrix} \\ &= b(1 - a^2) + a(b + ab) - a(-ab - b) \\ &= b - ba^2 + ab + a^2b + a^2b + ab \\ &= b + 2ab + a^2b \\ &= b(1 + 2a + a^2) \\ &= b(1 + a)^2 \end{aligned}$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}, x_i = \frac{|B_i|}{|M|}, B_i = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{b} \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$

$$\begin{aligned} |B_2| &= \begin{vmatrix} 1 & b & -a \\ -a & b & -a \\ -a & b & 1 \end{vmatrix} = 1 \begin{vmatrix} b & -a \\ b & 1 \end{vmatrix} - b \begin{vmatrix} -a & -a \\ -a & 1 \end{vmatrix} - a \begin{vmatrix} -a & b \\ -a & b \end{vmatrix} \\ &= (b + ab) - b(-a - a^2) - a(-ab + ab) \\ &= b + ab + ab + a^2b \\ &= b + 2ab + a^2b \\ &= b(1 + a)^2 \end{aligned}$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}, x_i = \frac{|B_i|}{|M|}, B_i = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{b} \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$

$$\begin{aligned} |B_3| &= \begin{vmatrix} 1 & -a & b \\ -a & 1 & b \\ -a & -a & b \end{vmatrix} = 1 \begin{vmatrix} 1 & b \\ -a & b \end{vmatrix} + a \begin{vmatrix} -a & b \\ -a & b \end{vmatrix} + b \begin{vmatrix} -a & 1 \\ -a & -a \end{vmatrix} \\ &= (b + ab) + a(-ab + ab) + b(a^2 + a) \\ &= b + ab + a^2b + ab \\ &= b + 2ab + a^2b \\ &= b(1 + a)^2 \end{aligned}$$

Cramer's Rule

$$M = \begin{bmatrix} 1 & -a & -a \\ -a & 1 & -a \\ -a & -a & 1 \end{bmatrix}, \vec{b} = \begin{bmatrix} b \\ b \\ b \end{bmatrix}, x_i = \frac{|B_i|}{|M|}, B_i = [\vec{a}_1 \quad \cdots \quad \vec{a}_{i-1} \quad \vec{b} \quad \vec{a}_{i+1} \quad \cdots \quad \vec{a}_n]$$

$$|M| = (1 + a)^2 (1 - 2a)$$

$$|B_1| = |B_2| = |B_3| = b(1 + a)^2$$

$$x_1 = x_2 = x_3 = \frac{b(1 + a)^2}{(1 + a)^2(1 - 2a)} = \left(\frac{b}{1 - 2a} \right), a \neq -1, \frac{1}{2}$$

Inverse by classical adjoint.

- Find a general form for the inverse of the following matrix:

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}$$

- **Recall:** $A^{-1} = |A|^{-1}C^T$, $C = [c_{ij}]$, $c_{ij} = (-1)^{(i+j)}|M_{ij}|$

Inverse by classical adjoint.

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}, A^{-1} = |A|^{-1}C^T, C = [c_{ij}], \quad c_{ij} = (-1)^{(i+j)}|M_{ij}|$$

$$\begin{aligned} |A| &= a \begin{vmatrix} 0 & b \\ b & b \end{vmatrix} - a \begin{vmatrix} a & b \\ 0 & b \end{vmatrix} + 0 \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} \\ &= a(0 - b^2) - a(ab - 0) + 0 \\ &= 0 - ab^2 - a^2b + 0 \\ &= -ab(a + b) \end{aligned}$$

$$|A| \neq 0 \rightarrow a, b \neq 0, a \neq -b$$

Inverse by classical adjoint.

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}, A^{-1} = |A|^{-1} C^T, C = [c_{ij}], \quad c_{ij} = (-1)^{(i+j)} |M_{ij}|$$

$$c_{11} = + \begin{vmatrix} 0 & b \\ b & b \end{vmatrix} = -b^2, c_{12} = - \begin{vmatrix} a & b \\ 0 & b \end{vmatrix} = -ab, c_{13} = + \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab$$

$$c_{21} = - \begin{vmatrix} a & 0 \\ b & b \end{vmatrix} = -ab, c_{22} = + \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab, c_{23} = - \begin{vmatrix} a & a \\ 0 & b \end{vmatrix} = -ab$$

$$c_{31} = + \begin{vmatrix} a & 0 \\ 0 & b \end{vmatrix} = ab, c_{32} = - \begin{vmatrix} a & 0 \\ a & b \end{vmatrix} = -ab, c_{33} = + \begin{vmatrix} a & a \\ a & 0 \end{vmatrix} = -a^2$$

Inverse by classical adjoint.

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}, A^{-1} = |A|^{-1}C^T, C = [c_{ij}], \quad c_{ij} = (-1)^{(i+j)}|M_{ij}|$$

$$|A| = -ab(a + b)$$

$$\begin{array}{lll} c_{11} = -b^2 & c_{12} = -ab & c_{13} = ab \\ c_{21} = -ab & c_{22} = ab & c_{23} = -ab \\ c_{31} = ab & c_{32} = -ab & c_{33} = -a^2 \end{array} \rightarrow C = \begin{bmatrix} -b^2 & -ab & ab \\ -ab & ab & -ab \\ ab & -ab & -a^2 \end{bmatrix}$$

$$A^{-1} = |A|^{-1}C^T = \left(\frac{1}{-ab(a + b)} \right) \begin{bmatrix} -b^2 & -ab & ab \\ -ab & ab & -ab \\ ab & -ab & -a^2 \end{bmatrix}$$

Inverse by classical adjoint.

$$A = \begin{bmatrix} a & a & 0 \\ a & 0 & b \\ 0 & b & b \end{bmatrix}$$

$$A^{-1} = \left(\frac{1}{-ab(a+b)} \right) \begin{bmatrix} -b^2 & -ab & ab \\ -ab & ab & -ab \\ ab & -ab & -a^2 \end{bmatrix}$$

$$A^{-1} = \left(\frac{1}{a+b} \right) \begin{bmatrix} b/a & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & a/b \end{bmatrix}, a, b \neq 0, a \neq -b$$